Sentence Classification Using Quantum Natural Language Processing and Comparison of Optimization Methods

K. M. M. Rajashekharaiah[1(✉)], Satyadhyan Chickerur[2], Goutam Hegde[1], Subrahmanya L. Bhat[1], and Shubham Annappa Sali[1]

[1] School of Computer Science and Engineering, K L E Technological University, Hubballi 580 031, Karnataka, India
kmmr@kletech.ac.in

[2] Centre for High Performance Computing, K L E Technological University, Hubballi 580 031, Karnataka, India
chickerursr@kletech.ac.in

Abstract. The natural language processing (NLP) demands high computing resourses to process language related problems. The evolution of quantum computing hardware and simulators create avenues to design and develop NLP methods to test on these platforms. This research uses quantum natural language processing (QNLP) to simulate the sentence classification and compares optimization techniques (Simultaneous perturbation stochastic approximation: SPSA and convergent optimization via most-promising-area stochastic search: COMPASS) for small dataset and simple sentence structures and the results are promising to show that with the COMPASS optimization technique the model performs better Table 1. When the ansatz depth is less and vice-versa. The research in quantum natural language processing is infancy, the simulation results are promising and the better results can be expected for large datasets with different sentence structures if real quantum computers are available in near future.

Keywords: Quantum Computing · Quantum Natural Language Processing (QNLP) · SPSA · COMPASS · Binary Classification · DISCOCAT Model

1 Introduction

Natural language processing (NLP) problems like automatic summarization, machine translation, information retrieval and sentiment analysis and few of these involve sentence classification, where they often require high computing resources to understand sentence meanings. The advent of quantum computing promises the computational speed up exponentially higher than the present classical computing. Then the apparent question is whether this model of computing

K L E Technological University.

© Springer Nature Switzerland AG 2023
D. Garg et al. (Eds.): IACC 2022, CCIS 1782, pp. 85–98, 2023.
https://doi.org/10.1007/978-3-031-35644-5_7

can also be applied to natural language processing. If this becomes true, such approach may be the end of finding a computing model for speed ups matching to address the language related problems.The Quantumness of computing field is infancy in research and the majority of the effort in this computing model is theoretical or its realization is carried out as simulation on classical platform.

The research in quantum computing led to the development of quantum computers accessible are termed Noisy Intermediate Scale Quantum (NISQ) devices, assuring some success in implementation of application in various domains like text analysis, cryptography, medicine, machine learning and cloud computing. The other aspects with this new computing paradigm, the information representation and encoding of information may direct to theoretical and realistic advances in representing and dealing out language related issues beyond computational speed ups. Motivated by theses view points, the quantum natural language processing (QNLP) a new paradigm of innovation targeted to develop the NLP models to run on quantum hardware (NISQ) or simulator. Most of the work in this area is theoretical and the experiments are classically simulated. However the recent work [7] shows the implementation of QNLP model on real quantum hardware in a very small scale.

This research follow the work [15] and tests the capabilities of the QNLP model by increasing the data set size and compare the results by using the optimization techniques such as SPSA (Simultaneous Perturbation Stochastic Approximation) and COMPASS minimization techniques with different depths of ansatz (discussed in next sections). The observations are the COMPASS optimization technique give good classification accuracy in comparison with SPSA for low depth of ansatz and vice versa for higher depth of ansatz. The content of this paper is structured as: the Sect. 2 covers the related work; in the Sect. 3 we discuss the basics of quantum computing and model of meaning with the help of DisCoCat model that helps to understand the how the quantum computing is native to NLP. The methodology is explained in Sect. 4, here the pipeline of information encoding, the circuit representation, and model training and optimization techniques discussed. The results are discussed in Sect. 5 for different optimization techniques and finally conclusions are drawn in Sect. 6.

2 Related Works

The QNLP is a field of research still in its early years; however the following references embody the knowledge of general quantum computing and quantum information to understand quantum environment and natural language processing. Nelson and Isaac [1] disseminated introduction to quantum computing and quantum information that prelude for researchers in this domain. Lee J O'Riordan et al., [2] presented the idea for encoding corpus to the quantum circuit model with reference to DisCoCat formalism which is convenient and native to how the quantum information is processed. The Johannes Bausch et al., [3] designed the quantum algorithm for finding the globally optimal parse with high success probability for context sensitivity of language models, which is quadratically faster

than classical methods very useful in NLP. Bob Coecke [4] discusses a very useful theoretical and arithmetical foundation for near term natural language processing with the starting point, the DisCoCat brand of natural language processing. Joachim Lambek [5] explains a computational algebraic approach to grammar.

Dea Bankova et al., [6] introduced a idea of entailment, exploiting ideas from the categorical semantics of partial know K L E Technological University ledge in quantum computing theory, this shows that entailment strength lifts compositionally the sentence level, giving a lower bound on sentence entailment. They explain the necessary properties of graded entailment such as continuity, and provide a process for calculating entailment strength. Konstantinos Meichanetzidis et al., [7] developed a QNLP model to demonstrate it is NISQ friendly and promises scalability as the feature of the quantum hardware improves over a time. Relying on the algebra of pregorups [8] proposed a mathematical framework with diagrammatic calculus which explains how the information flow among the words in a sentence in order to make up the meaning of complete sentence and which helps to compare the meaning of the words in the distributional model. Bharti et al., [9] explained many hybrid classical-quantum algorithms with Noisy Intermediate Scale Quantum (NISQ) technology which motivates to design algorithms for NISQ. The greater part of previously demonstrable quantum machine learning (QML) protocols are based on variational quantum circuit methods (VQC) (Benedetti et al., 2019) [10,11]. Ramesh and Vinay [12] present quantum speedups for string matching which is significant for language identification. Recently [7] implemented QNLP and optimized on a NISQ device the classifier with a small dataset size of 16 sentences. The current work presents the testing the performance of the QNLP model by increasing the data set size and compare the results by using the optimization techniques such as SPSA(Simultaneous Perturbation Stochastic Approximation) [15] and COMPASS (Convergent Optimization Via Most-Promising-Area Stochastic Search) minimization techniques with different depths of ansatz(discussed in next sections) [16].

3 Quantum Computing and Model of Meaning Framework

3.1 Quantum Computing Background

Giving the complete knowledge of quantum computing and quantum information is beyond the scope of this paper, the reader is directed to the text [1]. However, to be a self explainable manuscript, this section will describe the theory and the terminology behind quantum computing who had no experience of quantum computing and quantum information. The natural starting point is a "qubit" which is the basic component of storage unit in quantum computing and is analogous to a "bit" in classical computing. It is connected with a property of a physical system such as the roll of an electron ('up' or 'down' along some axis) or the polarization of a single photon in which the two states can be considered to be the vertical polarization and the horizontal polarization [18]. As classical

bit has a state - either 0 or 1 - a qubit also has a state. Two possible states for a qubit are the states $|0\rangle$ and $|1\rangle$, which as you might guess correspond to the states 0 and 1 for a classical bit. Notation like '$|\ \rangle$' is called the Dirac notation, and we'll be seeing it frequently, as it's the standard notation for states in quantum mechanics. The difference between bits and qubits is that a qubit can be in a state(ψ) other than $|0\rangle$ or $|1\rangle$. It is also possible to form linear combinations of states; it is called superposition [1]:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

The numbers α and β are complex numbers, even though for many purposes not much is lost by considering them as real numbers. The state of a qubit is also represented as a vector in a two-dimensional complex vector space. The states $|0\rangle$ and $|1\rangle$ are known as computational basis states, and form an orthonormal basis for this vector space. We can examine a bit to determine whether it is in the state 0 or 1. Instead, quantum mechanics tells us that we can only acquire much more restricted information about the quantum state. When we measure a qubit we get either the result 0, with probability $|\alpha|^2$, or the result 1, with probability $|\beta|^2$. The $|\alpha|^2 + |\beta|^2 = 1$, given that the probabilities must sum to one. Geometrically, we can infer this as the condition that the qubit's state be normalized to length 1. Thus, in general a qubit's state is a unit vector in a two-dimensional vector space.

Quantum theory is a probabilistic theory which predicts the probabilities for the outcomes of 0 and 1. An isolated qubit's evolution before being measured is explained by changing the state of the qubit by applying a unitary linear map U i.e. $|\psi'\rangle = \mathrm{U}|\psi\rangle$. Figure 1 represents such evolution.

The combined state space of the q qubit is given by tensor product of individual state spaces. So the dimension is 2^q. For two uncorrelated qubits $|\psi 1\rangle = \alpha 1|0\rangle + \beta 1|1\rangle$ and $|\psi 2\rangle = \alpha 2|0\rangle + \beta 2|1\rangle$ the joint space is [15]

$$|\psi 1\rangle \otimes |\psi 2\rangle = (\alpha 1\alpha 2, \alpha 1\beta 2, \beta 1\alpha 2, \beta 1\beta 2)$$

The combination of multiple qubits that interact with each other for a particular task is described by unitary map acting on the overall state space. The diagrammatic representation from Fig. 1 then extends also to a quantum circuit, such as the example shown in Fig. 2. Quantum circuit is a model for quantum computation, it consists of quantum gates. All the nouns appear as effects in the quantum circuit. Generally effects can be generated by using Rx gate or combination of Rx and Rz gate. The cups are replaces with bell state in quantum circuit. The bell effect is generated by combining a CNOT gate and a Hadamard gate. The triangle and parallel boxes represents the tensor products of states and unitary maps respectively. The sequential connections of boxes represents composition of linear maps. The quantum circuit is designed such that it will be a function of outcome probabilities. When the quantum circuit is run on the hardware, the qubits are measured. We have to run the circuits multiple times to obtain the statistics for the outcome probabilities. In order to obtain the result for a given problem, the design of the circuit has to encode the problem such

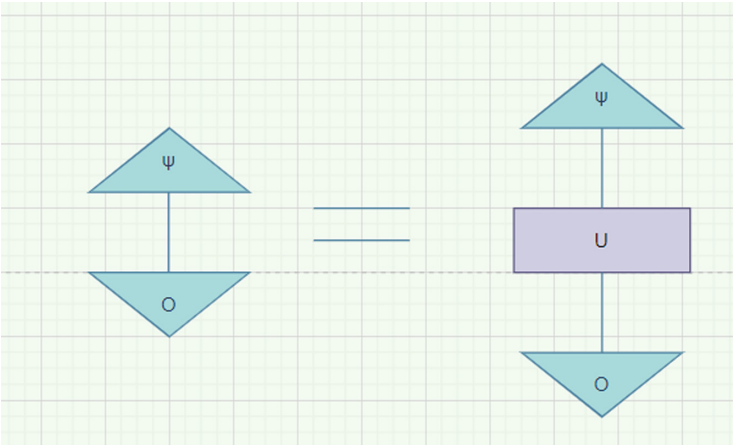


Fig. 1. Basic example of the diagrammatic representation, here of $\langle 0|\psi'\rangle = \langle 0|U|\psi'\rangle$, i.e. The evolution of a qubit in initial state $|\psi\rangle$ with unitary map U and then composed with the effect h⟨ 0 |, corresponding to the (non- deterministic) outcome '0'.

that that result is a function of the outcome probabilities. Hence, the selection of circuit is important [14].

The reason for high supremacy of quantum computing is quantum parallelism. We can create exponentially huge superpositions for the solution space of a problem. With more number of qubits, quantum computers supremacy increases exponentially. Adding extra transistors to a classical computer only increases power linearly; this is not the case here. In Quantum Computing there is another important property called as Entanglement. Here a pair's two individuals exist in a single quantum state. When one of the qubits' states is altered, the other one will instantly and predictably change as well.

The qubits are prone to noise and errors from the environment. A quantum computer that will be give advantage in solving large scale problems comes with more number of fault tolerant qubits. It will be helpful for solving some algorithms that is hard to be solved using classical computer. The currently available quantum computer known as noisy intermediate scale quantum computers comes with about 100 and is useful for development of theory and applications.

3.2 DisCoCat Model

The main model used in this work is DISCOCAT of Coecke et al. (2010). The complete discussion of DisCoCat model and how the quantum computing is native of NLP is beyond the scope of this research, however it directs to the text [4]. The fact that, the quantum computing environment and natural language share the make use of vector spaces for describing states albeit for very different reasons makes those two theories (essentially) coincide. Therefore we say that quantum natural language processing is the "quantum-native". What

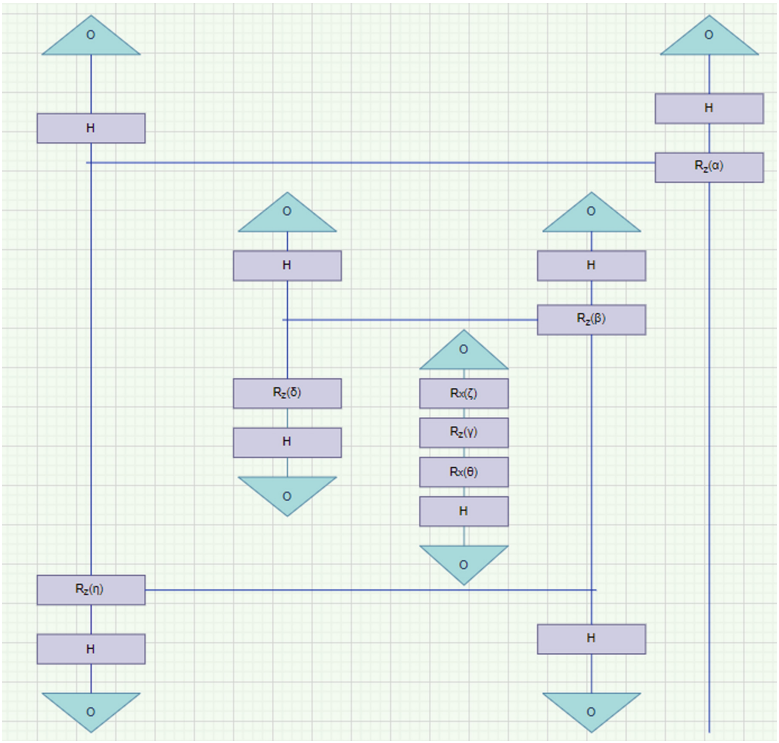


Fig. 2. The quantum circuit, which contains most of the gates related to this paper: the Hadamard H gate, the X-rotation gate Rx(β) by angle β, the controlled Z rotation gate, a component of which is a Z-rotation gate Rz(δ) by angle δ, and the quantum CNOT gate.

this implies is that natural language fit on quantum hardware, for the obvious reason that we have adopted a quantum model for language processing [4]. This is similar to simulation of quantum systems having been identified as a problem that would require a quantum computer [19]. But this of course hasn't stopped people from doing simulation of quantum systems on classical hardware. Category composition distribution semantics (also known as DisCoCat) uses category theory to combine the advantages of two very different approaches to linguistics: category grammars and distribution semantics. DisCoCat diagram represents the sentences in terms of string diagrams. The main reason for the use of DISCOCAT model in QNLP is that the underlying compact-closed structures provide the concept of Hilbert Space formulation of quantum theory. In this model the meaning of a word is expressed in a tensor, and the order is determined by the type of words expressed in a pregroup grammar. In this model a type p has left(pl) and right(pr) adjuncts and there are two reduction rules in the grammar [14].

$$p.p^l \rightarrow 1 \text{ and } p.p^r \rightarrow 1 \tag{1}$$

We use 'n' and 's' as atomic types to represent nouns and sentences respectively so type of transitive verb become $n^r.s.n^l$. For example sentence, Man prepares Dinner can be derived as

$$n.(n^l.s.n^r).n \rightarrow (n.n^r).s.(n^l.n) \rightarrow 1.s.1 \rightarrow s. \tag{2}$$

This indicates that the sentence is grammatically correct. The diagrammatic representation for the above sentence is



Fig. 3. Diagrammatic representation of a sentence

In Fig. 3, the "cups" ($\cup$) denote the grammar reductions. The conversion from pregroups to vector space semantics is achieved by a mapping F that sends atomic types to vector spaces (n to N and s to S) and composite types to tensor product spaces ($n^r.s.n^l$ to N$\bigotimes$S$\bigotimes$N). For example, a transitive verb becomes a tensor of order 3, which can be seen as a bilinear map N$\bigotimes$N $\rightarrow$ S, while an adjective (with type n.nl) can be seen as a matrix, representing a linear map N$\rightarrow$N. Further, F translates all grammar reductions to tensor contractions, so that the meaning of a sentence s = w1w2 ... wn with a pregroup derivation α is given by:

$$\mathrm{s} = \mathrm{F}(\alpha)[\mathrm{w1} \bigotimes \mathrm{w2} \bigotimes \ldots \bigotimes \mathrm{wn}] \tag{3}$$

Here, F turns α into a linear map that, apply to the tensor product of the word representations, by tensor-contracting that expression returns a vector for the complete sentence. As a concrete example, the meaning of the sentence "Man prepares Dinner" becomes s = m.P.d, where m, d $\in$ N and L $\in$ N $\bigotimes$ S $\bigotimes$ N. Above, P is a tensor of order 3, and s is a vector in S. Note that the underlying field of the vector spaces is not specified, and can, e.g., be R or C depending on the particular type of model (it is C in this work) [14]. Meaning computations like the instance above can be suitably represented by means of the diagrammatic calculus of compact closed categories [1], where the boxes refer to tensors, the order of which is determined by the number of their wires, while the cups are now the tensor contractions. Note the correspondence between the pregroup diag ram above with the one here, and how the grammatical derivation successfully dictates equally the shapes of the tensors and the contractions. These string diagrams logically are mapped to quantum circuits, as used in quantum computation [14] (Fig. 4).

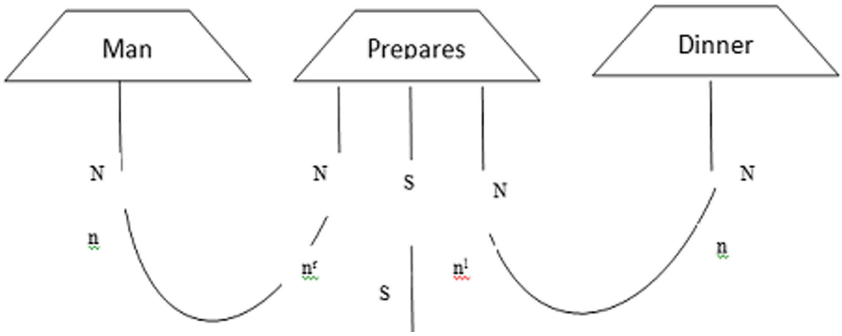


Fig. 4. Meaning computations as diagrammatic calculus of compact closed categories

4 Methodology

The previous section explained the quantum computing equivalent to classical programming involves the application of quantum gates on qubits according to a quantum circuit and the DisCoCat model. This section explains the methodology for sentence classification i.e. in particular the process of how to go from a sentence to its representation as a quantum circuit, on the basis of which the model predicts the label. Figure 2 presents the flow chart for the process of sentence classification using QNLP (Fig. 5).

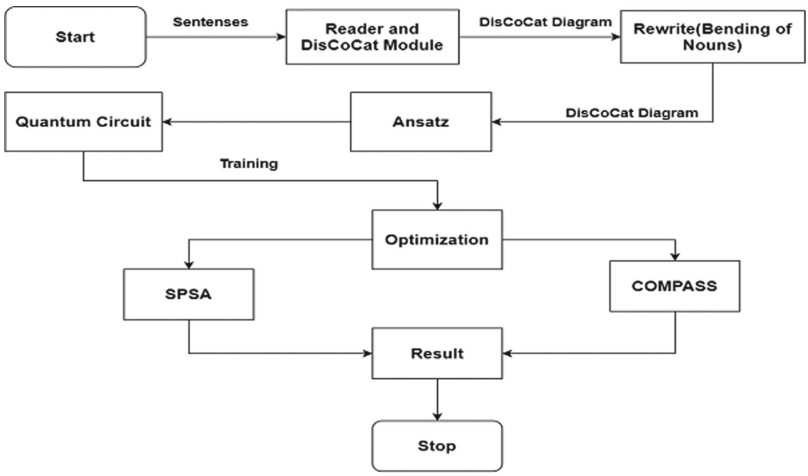


Fig. 5. Process pipeline of sentence classification in QNLP

The initial step is to obtain a syntax tree equivalent to the sentence, on the basis of which a DISCOCAT derivation is produced in a diagrammatic form.

To evade computational complications, this diagram is first optimized to yield the input into an ansatz that then determines the authentic translation into a quantum circuit [14,19]. The generated quantum circuit then executed on a simulator. These steps are described in more detail below.

Step 1: For the natural language experiments with large number of sentences (1000 and more) of different structures, the application of a pregroup parser to generate the syntax trees would be needed. However the current work considers the sentences with limited vocabulary and the small number of different grammatical structures. For example, with nouns, adjectives and transitive verbs having the respective types as n, $n.n^l$ and $n^r.n.n^l$, the sentence "person debugs useful software" is parsed as below:

$$\begin{aligned} &n.(n^r.s.n^l).(n.n^l).n \rightarrow \\ &(n.n^r).s.(n^l.n).(n^l.n) \rightarrow 1.s.1.1 \rightarrow s \end{aligned} \tag{4}$$

Step 2: Drawing the DISCOCAT diagram representing each word as a state and represent each reduction rule with cups. As discussed in the previous section, the following diagram represents linear-algebraic operations between tensors in vector spaces N, S and tensor products of them. For expediency, we also include the pregroup types as a reminder of the grammatical rules involved in each contraction. For the remainder of this paper, only the pregroup types will be shown since they are indicative of the vector spaces they are mapped to (Fig. 6).

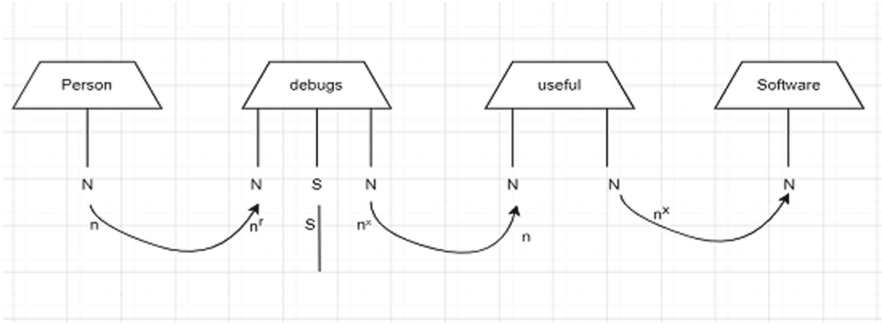


Fig. 6. DisCoCat representation of a sentence

Step 4: Mapping of DISCOCAT diagrams to quantum circuit. This is the step where abstract DisCoCat representation (Fig. 7) is mapped to specific quantum circuit. After the choice of ansatz (parameterization) every DisCoCat diagram is mapped to quantum circuit. The mapping of DisCoCat to quantum circuit is follows. Mapping of DisCoCat model to quantum circuit is determined by the choice of number of qubits qn and qs used to map n and s, choice of parameterized quantum and word states (effects) called ansatz. Each cup in DISCOCAT model is equivalent to Bell effect. Bell effect can be generated by using a CNOT gate

with a Hadamard gate. The number of qubits in the circuit is determined by the number of adjectives and transitive verbs in the sentences. Adjectives are states on 2 qubits and transitive verb are states on 3 qubits. The choice for number of qubits used for sentence $q_s = 1$. All the nouns appear as effects in the quantum circuit. These effects can be generated in two ways. First ways is to use two Rx gate and one Rz gate or we can use simply one Rx gate to generate qubit effects. Single parameter gives a more economical option that is considered. Therefore choice of parameter can be $p_n \in \{3, 1\}$. The next choice is consider the number of IQP (Instantaneous Quantum Polynomial [14] layers. For a circuit containing m qubits all are initialized to zero state. Followed by IQP layers, each of this IQP layer contains a Hadamard gate on each Qubit (To bring qubits to superposition or initial state) and m-1 controlled Rz gate connects adjacent qubits. The choice of IQP layers considered here are $d \in \{1, 2\}$ to keep the circuit depth small. So totally three hyperparameters are considered $(q_s, p_n, d) q_s = 1(fixed), p_n \in \{1, 3\}, d \in \{1, 2\}$. The total number of parameters is represented using k and it depends on size of vocabulary. The below figure Fig No 8: shows the quantum circuit corresponding to the DisCoCat diagram in Fig No 7.

Step 5: Model training and Optimization

The dataset used for the experiment consists of 150 sentences. The dataset were partitioned into 3 subsets namely training (90), test (35) and development (35) sets. The sentences used in the dataset are mainly of 4 grammatical structures. They are noun_tranisitive verb_noun, noun_transitive verb_adjective_noun, adjective_noun_transitive verb_noun, adjective_noun_transitive verb_adjective_noun. All the subsets are perfectly balanced with two classes (food and IT related). Every sentence in the dataset is represented by quantum circuit according to chosen ansatz. The output state of the quantum circuits can be represented as $l^i_\theta(P) := ||\langle i|P(\theta)\rangle|^2 - \varepsilon|$ where $i \in (0, 1)$ and ε is a small positive number and $l^0_\theta(P) + l^1_\theta(P) \leq 1$, so that $l_\theta(P) := (l^0_\theta(P), l^1_\theta(P))/(\sum_i l^i_\theta(P))$ defines probability distribution. The label predicted by the model is then obtained by rounding the results of probability distribution function. So that result will be either 0 or 1(food or IT). This is defined by $L_\theta(P) := \lfloor (l_\theta(P)) \rfloor$. The objective function used for model training is standard cross-entropy. If we assume L(P) the actual label according to the data. The cost function is given by $C(\theta) := \sum_{P \in T} L(P)^T \ .\log(l_\theta(P))$. Optimization algorithms are used for minimizing the cost function by $C(\theta)$ [15]. There are mainly two types of optimization techniques used in the experiment namely SPSA (Simultaneous Perturbation Stochastic Approximation) and COMPASS (Convergent Optimization Via Most-Promising-Area Stochastic Search).The results of both optimization techniques are compared for various parameters. Both the optimization algorithms are explained below.

Simultaneous perturbation stochastic approximation (SPSA)

SPSA is a method for minimizing differentiable multivariate functions. It is useful for those function for which calculating gradient is resource intensive or not possible. Similar to gradient descent SPSA is also iterative optimization

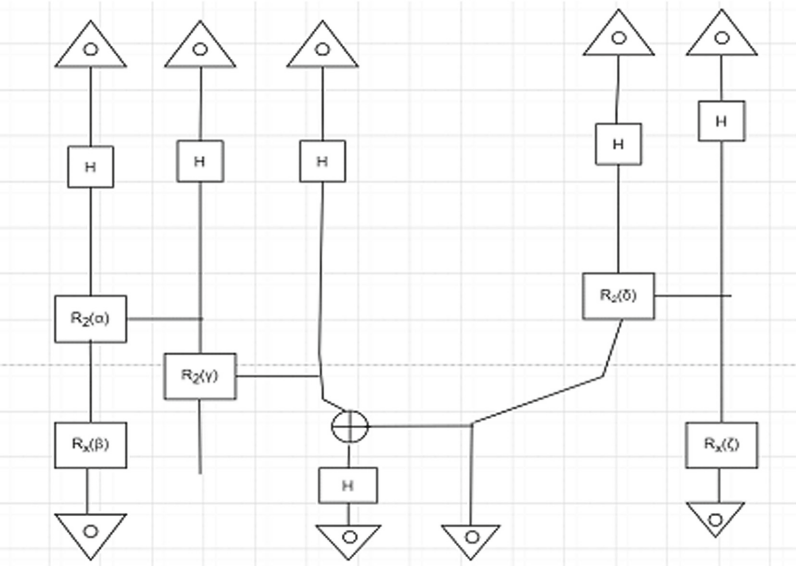


Fig. 7. Quantum circuit for a DisCoCat diagram

algorithm. The main feature of SPSA is that the objective function requires only two measurements regardless of the dimension of optimization problem. SPSA is most appropriate for high-dimensional problems where there is a requirement of determining many terms in the optimization process [13].

Compass Search Optimization (COMPASS) Compass (convergent optimization via most-promising-area stochastic search) is a optimization algorithm based on random search [16]. It is a search algorithm that does not use derivatives In this algorithm most promising region at each step is very adaptive which is chosen from a predetermined nested sets. Primarily all practical solutions are considered to be equally promising. Once some solutions have been visited, the promising index of every feasible solution is the sample mean performance of the visited solution that is closest to it. COMPASS solves a wide range of problems, including problems with very large numbers of feasible solutions, and works better than some of the existing algorithms COMPASS converges with probability 1 to a set of local optimal solutions and that it has robust finite-time performance. But the algorithm is not guaranteed to find a global minimum. It can be easily attracted to local minimum.

Step 6: Model testing After training the model, it will be used for predicting the labels for test dataset. Then the accuracy is calculated for the test data and the graphs for accuracy and cost are drawn.

5 Results and Discussions

The simulation is conducted on both Quantum IBM experience and Google colab. The libraries used for experiments are, Lambeq, Pytket (python module for interfacing with CQC (Cambridge Quantum Computing)) a set of quantum

Table 1. The Simulation results of SPSA and COMPASS optimization Techniques. The graphs for both optimization algorithms with different variations of parameters are shown in the Fig. 8. For the results obtained as mentioned in the Table 1.

Ansatz (Qs, Pn, d)	Classification Accuracy with SPSA	Classification Accuracy with COMPASS
(1, 1, 1)	0.633334	0.86667
(1, 1, 2)	0.966667	0.93334
(1, 3, 1)	0.533336	1.0
(1, 3, 2)	1.0	0.93334



Fig. 8. The performance of model optimization Techniques SPSA and COMPASS with various parameter.

computing programming tools. The python library Noisyopt used for optimizing noisy functions. SPSA and COMPASS from Noisyopt and other supporting tools. The Table 1, shows the simulation results. The table shows that the Compass

optimization algorithm gives good results for the quantum circuits with depth = 1. SPSA optimization algorithm gives good results for the circuits with depth = 2. Compass is not robust to noise but gives good results for the quantum circuits with minimum depth. SPSA is robust to noise so noises are less likely to hinder the optimization process. Quantum circuits with larger depth (large number of qubits) contains more noise and the dimension of the optimization problem is also large in this case. As SPSA is robust to noise and gives good results for high dimensional optimization problem, the SPSA algorithm gives high accuracy for quantum circuits with more depth.

6 Conclusion

In this work we have simulated the sentence classification using QNLP on classical system with moderate data set size. This work also compares two machine learning model optimization techniques namely SPSA and COMPASS. The classification accuracy obtained by using SPSA and COMPASS for the parameters $(1, 1, 1), (1, 1, 2), (1, 3, 1), (1, 3, 2)$ are 0.63334, 0.96667, 0.53336, 1.0 and 0.86667, 0.93334, 1.0, 0.93334 respectively. The results are promising to show that with the COMPASS optimization technique the model performs better Table 1 when the ansatz depth is less and vice-versa if the optimization technique is SPSA Table 1. The research in QNLP is infancy and the implementations can be realized with large data set of different sentence format if the real quantum computer is available in the near feature.

References

1. Nielsen, M.A., Chuang, I.L.: Quantum Computation and Quantum Information: 10th Anniversary Edition, 10th edn. Cambridge University Press, New York (2011)
2. O'Riordan, L.J., Doyle, M., Baruffa, F., Kannan, V.: A hybrid classical-quantum workflow for natural language processing. Mach. Learn.: Sci. Technol. **2**(1), 015011 (2020)
3. Bausch, J., Subramanian, S., Piddock, S.: A quantum search decoder for natural language processing (2020)
4. Coecke, B., de Felice, G., Meichanetzidis, K., Toumi, A.: Foundations for near-term quantum natural language processing (2020)
5. Lambek, J.: From Word to Sentence. Polimetrica, Milan (2008)
6. Bankova, D., Coecke, B., Lewis, M., Marsden, D.: Graded entailment for compositional distributional semantics. J. Lang. Modell. **6**(2), 225–260 (2019)
7. Meichanetzidis, K., Toumi, A., de Felice, G., Coecke, B.: Grammar-aware question-answering on quantum computers (2020)
8. Coecke, B., Sadrzadeh, M., Clark, S.: Mathematical foundations for a compositional distributional model of meaning. Linguist. Anal. **36**, 345–384 (2010)
9. Bharti, K., et al.: Noisy intermediate-scale quantum (NISQ) algorithms (2021)
10. Benedetti, M., Lloyd, E., Sack, S., Fiorentini, M.: Parameterized Quantum Circuits as Machine Learning Models. Quantum Sci. Technol. **4**(4), 043001 (2019)

11. Rajashekharaiah, K.M.M., et al.: Transfer learning using variational quantum circuits. In: Garg, D., Jagannathan, S., Gupta, A., Garg, L., Gupta, S. (eds.) IACC 2021. Communications in Computer and Information Science, vol. 1528, pp. 254–267. Springer, Cham (2022). https://doi.org/10.1007/978-3-030-95502-1_20
12. Ramesh, H., Vinay, V.: String matching in O(n+m) quantum time. J. Discrete Algorithms **1**(1), 103–110 (2003). Combinatorial Algorithms
13. Spall, J.C.: Implementation of the simultaneous perturbation algorithm for stochastic optimization. IEEE Trans. Aerosp. Electron. Syst. **34**(3), 817–823 (1998)
14. Lorenz,R., Pearson, A., Meichanetzidis, K., Kartsaklis, D., Coecke, B.: QNLP in practice: running compositional models of meaning on a quantum computer. arXiv:2102.12846 [cs.CL] (2021)
15. Spall, J.C.: Implementation of the simultaneous perturbation algorithm for stochastic optimization. IEEE Trans. Aerosp. Electron. Syst. **34**(3), 817–823 (1988)
16. Hong, L.J., Nelson, B.L.: Discrete optimization via simulation using COMPASS. Oper. Res. **54**(1), 115–129 (2006)
17. https://en.wikipedia.org/wiki/Qubit
18. Feynman, R.P.: Simulating physics with computers. Int. J. Theor. Phys. **21**(6/7), 467 (1982)
19. Grefenstette, E., Sadrzadeh, M.: Experimental support for a categorical compositional distributional model of meaning. In: Proceedings of the Conference on Empirical Methods in Natural Language processing, pp. 1394–1404. Association for Computational Linguistics (2011)